Likelihood targets of a nested ensemble Kalman filter

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Abstract

State-space models with a static parameter have an observed-data likelihood that integrates out a latent Markov process. Nested particle filters for the parameter become expensive as the observation dimension grows. An ensemble Kalman forecast supplies a cheaper likelihood increment, and a nearest-neighbour prediction of that increment is cheaper still. These replacements change the posterior in different ways. We separate the two errors. The plain nested filter targets the posterior induced by the ensemble Kalman likelihood. The nearest-neighbour weight targets the posterior induced by the surrogate. A nearest-neighbour screen that is then corrected by the ensemble Kalman likelihood leaves the first of those posteriors invariant. A Rao-Blackwellised weight restores the true observation density whilst the forecast remains Gaussian. When the observation density is already that linear Gaussian density, the two weights agree, so the correction has nothing to change. On a Lotka–Volterra diffusion whose prey observation has variance equal to the latent prey, the plain filter lies 0.0586 and 0.0354 below bootstrap sequential Monte Carlo squared on the log predation coefficient and the log predator death rate. Neither gap exceeds its standard error. A uniform bound between the surrogate posterior and the ensemble Kalman posterior is not obtained.

Keywords: Lotka–Volterra model, Rao-Blackwellisation, sequential Monte Carlo, state-space model, surrogate likelihood

1 Introduction

State-space models are a standard description of a latent Markov process observed with noise. A static parameter governs the transition, the observation, or both. Applications include predator–prey dynamics, epidemic incidence and systems with many observed components (Golightly et al. 2026; Temfack and Wyse 2026). The inferential target at time t is the posterior distribution of the static parameter given the observations so far. The observed-data likelihood that enters that posterior is an integral over the latent path, and the integral is rarely available in closed form.

Sequential Monte Carlo squared (SMC2) propagates a cloud of parameter particles. Each particle carries a particle filter, the particle filter supplies an estimate of the likelihood increment, and a particle Metropolis–Hastings kernel moves the parameter particles when the effective sample size falls (Chopin et al. 2012). The estimate is unbiased for the true increment, and the parameter marginal of the extended target is the exact posterior for any fixed number of state particles. The variance of the estimate grows with the length of the series and, more sharply, with the observation dimension. In that regime a particle filter can require a number of state particles that is prohibitive.

The ensemble Kalman filter replaces the forecast distribution by a Gaussian distribution whose mean and covariance are the sample moments of an ensemble, and it updates the ensemble with a Kalman shift. The likelihood contribution is the Gaussian marginal of that forecast. The contribution is exact for the Gaussian approximation and is biased for the true predictive density

of the state-space model. Drovandi et al. (2022) place the contribution inside a Metropolis–Hastings kernel on a fixed data set. Golightly et al. (2026) place it inside the SMC2 recursion and call the resulting scheme the nested ensemble Kalman filter. The parameter target is the posterior induced by the ensemble Kalman likelihood, which we denote $\hat{\pi}_{\mathrm{enkf}}$, rather than the exact posterior π_t. For a nonlinear or non-Gaussian observation the same construction first replaces the observation density by a linear Gaussian approximation. A Rao-Blackwellised weight then puts the true observation density back into the increment, whilst the forecast remains Gaussian, and targets a third posterior, $\tilde{\pi}_{\mathrm{enkf}}$.

On the Lotka–Volterra diffusion in Section 5.2 of Golightly et al. (2026), prey are observed under a Gaussian density whose variance equals the latent prey, and predators are latent. That paper reports that the plain filter shifts the posterior means of the predation coefficient and the predator death rate downward, because the observation variance is replaced by a function of the forecast mean, and that the Rao-Blackwellised filter reduces the discrepancy at greater cost. The comparison uses 1000 parameter particles, an ensemble whose size is allowed to grow, delayed acceptance, 50 replicates and a long particle Markov chain Monte Carlo run as the reference. The same paper uses a nearest-neighbour average as a screen inside delayed acceptance, so that many proposals avoid an ensemble Kalman likelihood. The discussion lists a further use of that average: as the incremental weight itself, at a fixed ensemble size, which would give a cheaper filter at the expense of accuracy. The posterior induced by that weight is not defined there, and no distance between it and $\hat{\pi}_{\mathrm{enkf}}$ is given.

Those omissions matter, because the two replacements are different errors. The first is the ensemble Kalman likelihood against the true likelihood, and hence $\hat{\pi}_{\mathrm{enkf}}$ against π_t. The second is a nearest-neighbour prediction against the ensemble Kalman likelihood. A weight that uses the prediction targets a further posterior, which we denote π_{knn}. A screen that is corrected by the ensemble Kalman likelihood leaves $\hat{\pi}_{\mathrm{enkf}}$ invariant. A comparison that scores π_{knn} against π_t, or against a gold-standard run for π_t, attributes both errors to the ensemble Kalman approximation. The Rao-Blackwellised weight sits in between, in a precise sense. It removes the linear Gaussian approximation of the observation and retains the Gaussian forecast, so $\tilde{\pi}_{\mathrm{enkf}}$ is not π_t. When the true observation density is already the linear Gaussian density inside the filter, the ratio that is meant to correct the observation equals one, and the Rao-Blackwellised weight reproduces the ensemble Kalman weight. Golightly et al. (2026) compare the two weights in examples where the observation is not linear and Gaussian. The identity says that the comparison is informative in that case, and that the correction has no inferential content when the observation model inside the filter is already the right one.

Our contribution is to separate the errors and to say which posterior each modification targets.

(i) We name the four targets and keep them distinct. Bootstrap SMC2 targets π_t. The plain nested filter targets $\hat{\pi}_{\mathrm{enkf}}$. The Rao-Blackwellised weight targets $\tilde{\pi}_{\mathrm{enkf}}$. The nearest-neighbour weight targets π_{knn}. Figure 1 is the corresponding diagram.

(ii) We record the delayed-acceptance argument that leaves $\hat{\pi}_{\mathrm{enkf}}$ invariant when the neighbour average is only a screen, and the elementary total-variation bound that would control π_{knn} if the two log-likelihoods differed by a uniform constant. The hypothesis of that bound is not met. The ensemble Kalman log-likelihood is redrawn with the ensemble, and the diagnostic we compute is not such a constant.

(iii) We prove that the Rao-Blackwellised increment equals the ensemble Kalman increment when the two observation densities agree. The identity is specific to that case. A linear Gaussian illustration with a deliberately wrong observation variance shows that the correction can move the posterior means when the densities differ.

(iv) On the Lotka–Volterra observation we compare the plain filter, the Rao-Blackwellised filter and bootstrap SMC2. The design is smaller than Section 5.2 of Golightly et al. (2026),

and the reference is SMC2 itself. The downward shift in the two log rate parameters lies inside its standard error.

The numerical sections state every setting that differs from Golightly et al. (2026), including the shape-rate reading of the gamma priors and the treatment of non-positive Euler coordinates. They are not a reproduction of the tables in that paper. We do not analyse an epidemic series. Temfack and Wyse (2026) already study a nested ensemble scheme for incidence, and they are explicit that a Gaussian assumption remains.

The remainder of this paper is organised as follows. Section 2 describes the neighbouring algorithms and which posterior each one targets. Section 3 defines the two errors, states the identity, and records the total-variation calculation that does not close. Section 4 reports the linear Gaussian checks and the Lotka–Volterra comparison. Section 5 discusses what those comparisons support and what remains open.

2 Related work

The algorithms in this section all propagate information about a static parameter of a state-space model, or they are the constructions with which the nested filter is easily confused. They do not share a parameter target. Table 1 records the distinction that the rest of the paper uses. The nearest-neighbour row is the target defined in Section 3. The other rows are the algorithms as published.

Table 1: Parameter targets of the neighbouring schemes. The exact posterior is π_t. The ensemble Kalman posterior is the posterior induced by the ensemble Kalman likelihood. Citations for the rows are Chopin et al. (2012), Drovandi et al. (2022), Katzfuss et al. (2018), Golightly et al. (2026) and Temfack and Wyse (2026).

Method	Weight	Parameter target	Sequential
SMC2	Unbiased particle estimate of the increment	Exact posterior π_t	Yes
Ensemble MCMC	Ensemble Kalman likelihood	Ensemble Kalman posterior, fixed series	No
Particle ensemble Kalman filter	Integrated ensemble Kalman likelihood	Ensemble Kalman posterior	Yes
Plain nested filter	Ensemble Kalman increment	$\hat{\pi}_{\mathrm{enkf}}$	Yes
Rao-Blackwellised weight	Gaussian forecast, true observation density restored	$\tilde{\pi}_{\mathrm{enkf}}$	Yes
Nearest-neighbour weight	Prediction of the ensemble Kalman increment	π_{knn}	Yes
Ensemble SMC2	Unbiased Gaussian predictive density	Posterior under that Gaussian model	Yes

SMC2 is the scheme whose parameter marginal is exact. Chopin et al. (2012, Proposition 1) write an extended target that includes the particle-filter randomness, and whose θ-marginal is π_t for any fixed number of state particles. The argument uses unbiasedness of the product of incremental weights for the true likelihood. Replacing the inner particle filter by an ensemble Kalman filter removes that unbiasedness. The marginal that remains is $\hat{\pi}_{\mathrm{enkf}}$, not π_t. The same proposition therefore gives no distance between $\hat{\pi}_{\mathrm{enkf}}$ and π_t, and no distance between π_{knn} and $\hat{\pi}_{\mathrm{enkf}}$. The nearest-neighbour weight is not unbiased for the ensemble Kalman likelihood either.

Ensemble Markov chain Monte Carlo places the ensemble Kalman likelihood in the acceptance ratio of one chain, run on a fixed series (Drovandi et al. 2022). One rejuvenation sweep

of the nested filter is one iteration of that scheme, but the chain itself is not sequential, and it does not update a posterior as observations arrive. Even when the forecast is exactly Gaussian, the plug-in Gaussian density is a biased estimator of the idealised Gaussian predictive that uses the true forecast mean and covariance. Drovandi et al. (2022, Section 3.3) discuss an unbiased estimator of that Gaussian density, of Ghurye–Olkin type, available when the forecast is Gaussian and the ensemble is large enough. Temfack and Wyse (2026) use the same style of estimator for an epidemic model with a state-dependent observation variance. An estimator of that kind removes the finite-sample bias of the plug-in Gaussian density. It does not remove the error that separates $\hat{\pi}_{\mathrm{enkf}}$ from π_t, because the idealised Gaussian predictive is already an approximation to the true predictive. Temfack and Wyse (2026) are explicit that the Gaussian assumption remains. Their mutation is particle marginal Metropolis–Hastings. They do not use the Rao-Blackwellised weight of Section 3.

A different sequential construction gives the parameter an artificial transition and resamples it with a shrinkage kernel, so that the filter does not collapse onto a single parameter value (Liu and West 2001). The particle ensemble Kalman filter of Katzfuss et al. (2018) weights parameter particles by an integrated ensemble Kalman likelihood and, for a constant parameter, needs a kernel of that kind to keep the parameter cloud alive. The nested filter keeps θ static inside each rejuvenation and restores diversity by a Metropolis–Hastings move under the likelihood the weight is using. Katzfuss et al. (2018, Proposition 2) gives the variance of the log ensemble Kalman likelihood as being of order n/N in one linear example, as the observation dimension n and the ensemble size N grow, when the forecast members are independent draws from the true forecast. The ensemble inside the nested filter is interacting, so the proposition is an analogy for the cost of controlling that variance. It is not a uniform bound on the log-likelihood, and it is not a bound on π_{knn}.

The weighted ensemble Kalman filter proposes a state ensemble with a Kalman shift and then reweights it (Papadakis et al. 2010). The weight contains the observation density and the transition density. Papadakis et al. (2010) note that the transition density is usually unavailable when the dynamical time step is finer than the observation interval, which is the regime of the diffusions in Section 4. The Rao-Blackwellised weight in Section 3 is built so that the Gaussian forecast appears in the target and in the proposal and therefore cancels. The transition density between observation times is never evaluated. The weighted ensemble Kalman filter is a state filter. It does not supply a posterior for a static parameter, and its weight is not evidence for or against the ordering of $\hat{\pi}_{\mathrm{enkf}}$ and $\tilde{\pi}_{\mathrm{enkf}}$.

Delayed acceptance screens a proposal with a cheap approximation and then corrects the screen with the intended target (Christen and Fox 2005). If the parent kernel is irreducible for that target, the proposal is reversible, and the approximation is positive wherever the proposal is positive, the target remains invariant. Golightly et al. (2026, Section 4.2) use this construction with an inverse-distance neighbour average in place of the ensemble Kalman likelihood at the first stage, and with the ensemble Kalman likelihood at the second stage. The invariant distribution is $\hat{\pi}_{\mathrm{enkf}}$. Sherlock et al. (2016, Section 5.1) contrast that use with algorithms that take the approximation as the target: draws aimed at the surrogate are not draws from the expensive posterior. Bon et al. (2021) put a surrogate inside the mutation of a sequential Monte Carlo sampler and keep the expensive likelihood as the second-stage target. In both cases the second stage is what protects the intended posterior. The nearest-neighbour weight in Section 3 is the use those papers set aside. It targets π_{knn}.

The gap left by this literature is specific. Golightly et al. (2026) define $\hat{\pi}_{\mathrm{enkf}}$ and $\tilde{\pi}_{\mathrm{enkf}}$, compare them empirically, and use the neighbour average as a screen. They do not define the posterior induced by using that average as the weight, they do not separate that error from the ensemble Kalman error, and they do not record when the Rao-Blackwellised weight collapses to the plain weight. The unbiased Gaussian correction in Drovandi et al. (2022) and Temfack and Wyse (2026) addresses a third error, the plug-in bias of a Gaussian density, and leaves both of

the errors in Figure 1 untouched. Section 3 treats the two errors that the nested filter can make. Section 4 compares the targets without adding the errors together.

3 Methods

We consider a state-space model with static parameter $\theta \in \Omega \subseteq \mathbb{R}^{d_\theta}$. The latent state $X_t \in \mathbb{R}^{d_x}$ has initial density $p_0^{(\theta)}$ and transition densities $p_t^{(\theta)}(x_t \mid x_{t-1})$ for $t = 1, \ldots, T$. The observation $Y_t \in \mathbb{R}^{d_y}$ has density $f_t^{(\theta)}(y_t \mid x_t)$. The exact parameter posterior at time t is

$$\pi_t(\theta \mid y_{0:t}) \propto \pi(\theta)\, p_t^{(\theta)}(y_{0:t}), \tag{1}$$

where $p_t^{(\theta)}(y_{0:t})$ is the observed-data likelihood. The sequential factorisation of that likelihood multiplies the posterior at time $t-1$ by the predictive density of the new observation. That predictive density is the integral of the transition and the observation density against the filtering distribution of the state. It is the object the constructions below replace.

3.1 Ensemble Kalman likelihood

The linear Gaussian observation model is

$$f_t^{(\theta)}(y_t \mid x_t) = \mathcal{N}(y_t; Hx_t, R). \tag{2}$$

An ensemble of size N represents the filtering distribution at the previous time. Algorithm 1 is one ensemble Kalman step at a fixed θ, in the stochastic form used by Golightly et al. (2026, Algorithm 3). The sample covariance uses the divisor $N-1$.

Algorithm 1. *Ensemble Kalman step at time t, for one parameter.*

1. Forecast. For $j = 1, \ldots, N$, draw $\tilde{x}_t^j$ from the transition $p_t^{(\theta)}(\cdot \mid x_{t-1}^j)$.
2. Forecast moments. Set $\hat{\mu}_{t|t-1} = N^{-1} \sum_{j=1}^N \tilde{x}_t^j$, and let $\hat{\Sigma}_{t|t-1}$ be the sample covariance of the forecast ensemble.
3. Likelihood contribution. Evaluate
$$\hat{p}_{\mathrm{enkf},t}^{(\theta)}(y_t \mid y_{0:t-1}) = \mathcal{N}\big(y_t; H\hat{\mu}_{t|t-1},\, H\hat{\Sigma}_{t|t-1}H^\top + R\big). \tag{3}$$
4. Kalman gain. Set $\hat{K}_t = \hat{\Sigma}_{t|t-1}H^\top (H\hat{\Sigma}_{t|t-1}H^\top + R)^{-1}$.
5. Analysis. For each member, draw a pseudo-observation $\tilde{y}_t^j \sim \mathcal{N}(H\tilde{x}_t^j, R)$ and set $x_t^j = \tilde{x}_t^j + \hat{K}_t(y_t - \tilde{y}_t^j)$.

Equation (3) is exact for the Gaussian forecast. It is a biased estimator of the true predictive density. The product of these contributions, multiplied by the prior, defines a parameter posterior that still depends on the ensembles drawn along the series. Golightly et al. (2026, Section 4.1) write the corresponding extended target. Its θ-marginal is $\hat{\pi}_{\mathrm{enkf},t}$. The first error is the difference between $\hat{\pi}_{\mathrm{enkf},t}$ and $\pi_t(\theta \mid y_{0:t})$.

3.2 Plain nested filter

The outer recursion follows Golightly et al. (2026, Algorithm 6). At time t, each of M parameter particles is updated by Algorithm 1 and its weight is multiplied by (3). Let $\gamma \in (0, 1)$ be the effective-sample-size fraction. If the effective sample size of the updated cloud is at least γM, the cloud is left as it is. Otherwise the particles are resampled with replacement, the weights

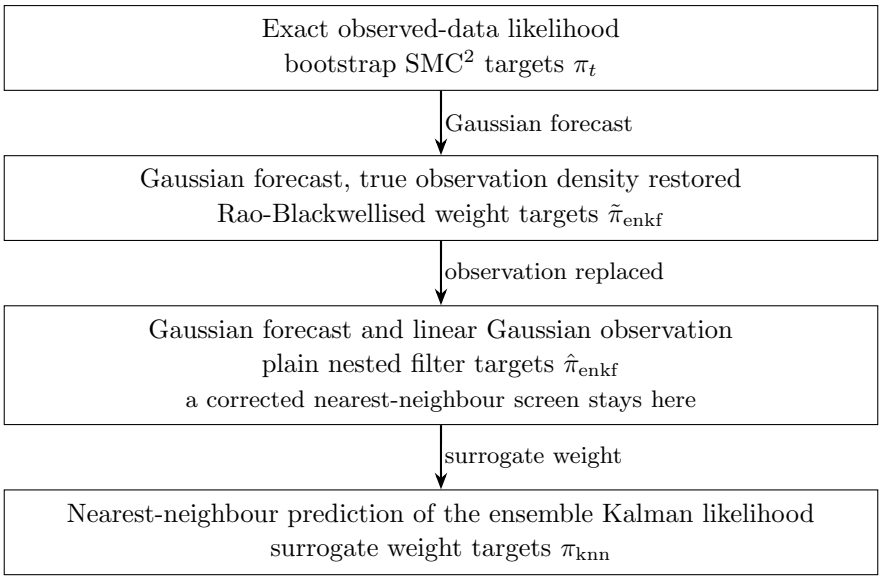


Figure 1: Parameter targets induced by four likelihoods. Each arrow replaces the likelihood used for the parameter weight. The first error is the ensemble Kalman likelihood against the true likelihood: the forecast remains Gaussian, so $\tilde{\pi}_{\text{enkf}}$ and $\hat{\pi}_{\text{enkf}}$ both differ from π_t. The second error is the nearest-neighbour prediction against the ensemble Kalman likelihood. A nearest-neighbour screen that is corrected by the ensemble Kalman likelihood stays on $\hat{\pi}_{\text{enkf}}$.

are reset to $1/M$, and each particle is offered one Metropolis–Hastings proposal. The proposal is a random walk whose covariance is ζ times the sample covariance of the resampled particles other than the particle being moved. We take $\zeta^2 = 2.56^2/d_\theta$, the scaling Golightly et al. (2026, Section 4.1) cite from the pseudo-marginal random-walk literature. The acceptance ratio uses the prior and the ensemble Kalman likelihood of the series so far. The likelihood at a proposed value is obtained by rerunning Algorithm 1 from the known initial condition. For this random walk the Gaussian proposal is symmetric, provided the covariance is held fixed during the ratio, and the proposal ratio equals one.

The printed Algorithm 6 of Golightly et al. (2026) places the effective-sample-size test inside the loop over parameter particles. The text of their Section 4.1 applies one resample-move step when the effective sample size of the updated cloud falls below γM, which is the order SMC2 uses. We follow that text: every particle receives its likelihood increment, and the test is applied once.

Golightly et al. (2026, Section 4.1) also allow the ensemble size to grow when an estimate of the variance of the log ensemble Kalman likelihood exceeds a threshold. The comparisons in Section 4 hold N fixed. A fixed ensemble size is the regime in which their discussion suggests a surrogate weight. Holding N fixed does not change the nominal target of a Metropolis–Hastings kernel that evaluates (3). It removes the extra randomness of the growth rule from the comparison.

For the nonlinear Gaussian observation in Section 4.2, the matrix R in (3) is replaced by the scalar $\tilde{R} = H\hat{\mu}_{t|t-1}$, as in Section 5.2 of Golightly et al. (2026). The plain filter then targets $\hat{\pi}_{\text{enkf}}$ under that choice. It does not evaluate the true observation density. Figure 1 places this target relative to the other three.

3.3 Rao-Blackwellised weight

For a general observation density f, the filter can still be run under a linear Gaussian approximation $g_t^{(\theta)}(y_t \mid x_t) = \mathcal{N}(y_t; \tilde{H}x_t, \tilde{R})$. One ensemble Kalman step under g produces an analysis

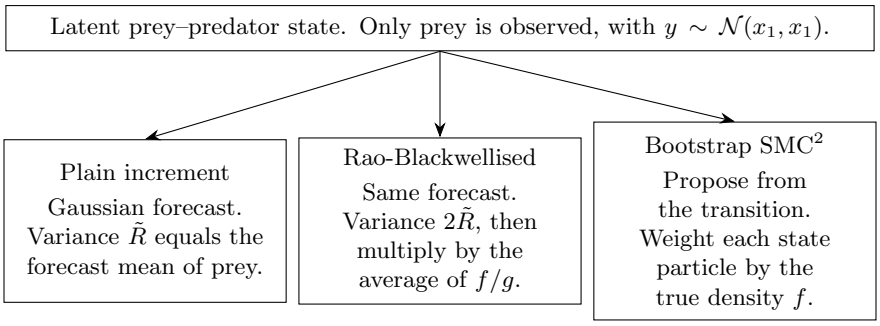


Figure 2: One observation update for the Lotka–Volterra model of Section 4.2. The plain increment replaces the state-dependent variance x_1 by the forecast mean of the prey. The Rao-Blackwellised increment inflates that variance and then reintroduces f through f/g. Bootstrap SMC2 uses f directly. The nearest-neighbour increment replaces the plain increment by an inverse-distance average of stored plain increments and does not evaluate a forecast at the particle being weighted.

ensemble and a Gaussian predictive likelihood $\hat{g}^{(\theta)}_{\mathrm{enkf},t}$. The Rao-Blackwellised increment is the average, over the N analysis members, of

$$\hat{g}^{(\theta)}_{\mathrm{enkf},t}(y_t)\,\frac{f^{(\theta)}_t(y_t \mid x^j_t)}{g^{(\theta)}_t(y_t \mid x^j_t)}. \tag{4}$$

The parameter target induced by these increments is $\tilde{\pi}_{\mathrm{enkf},t}$. The ratio f/g puts the true observation density back. The Gaussian forecast remains, so the transition density between observation times is never required, and $\tilde{\pi}_{\mathrm{enkf},t}$ is not π_t. Figure 2 shows the three increments on the observation used in Section 4.2.

Lemma 1. *If $f^{(\theta)}_t = g^{(\theta)}_t$, the Rao-Blackwellised increment equals the ensemble Kalman increment* (3).

Proof. The ratio in (4) equals 1 for every analysis member. The average of those terms is the Gaussian predictive likelihood under g. Under the hypothesis, that likelihood is (3). □

The incremental weights therefore agree, and so do the parameter targets. Evaluating the ratio costs an extra density for each ensemble member and does not change the estimate. The cancellation uses equality of the two observation densities. It does not, by itself, describe a nonlinear observation, and it does not say that $\hat{\pi}_{\mathrm{enkf}}$ equals π_t. Both weights can be wrong together when the forecast is replaced by a Gaussian.

In the Lotka–Volterra comparison the true density f is Gaussian with variance equal to the prey component. The plain filter uses $\tilde{R} = H\hat{\mu}_{t|t-1}$. The Rao-Blackwellised filter uses the same quantity multiplied by 2, which is the inflation in Section 5.2 of Golightly et al. (2026), and it evaluates f/g at the analysis members. The inflation is a proposal adjustment. It is not a claim that the true variance is twice the forecast mean. Bootstrap SMC2 does not use g. Its state weight is f, its proposal is the transition, and its parameter target is π_t, up to the discretisation of that transition.

3.4 Nearest-neighbour weight

Golightly et al. (2026, Section 4.2) build a surrogate from stored log ensemble Kalman likelihoods. For a point θ, let $\theta^{[1]}, \ldots, \theta^{[k]}$ be the k nearest neighbours under a chosen metric, with distances $d_{[i]}$. The surrogate log-likelihood is the inverse-distance average of the stored values. If some distance is zero, the surrogate equals the stored log-likelihood at that neighbour.

In delayed acceptance this average replaces the ensemble Kalman likelihood in the first-stage acceptance probability only. The second stage evaluates the ensemble Kalman likelihood. Under the conditions of Christen and Fox (2005, Theorem 1), recalled in Section 2, the invariant distribution remains $\hat{\pi}_{\text{enkf}}$.

The second error begins when the same average is used as the incremental weight, with no second stage. Write $\ell_{\text{knn},t}(\theta)$ for the surrogate log-likelihood of $y_{0:t}$ at θ, built from a stored cloud. The weights then target

$$\pi_{\text{knn},t}(\theta) \propto \pi(\theta)\, \exp\bigl(\ell_{\text{knn},t}(\theta)\bigr), \tag{5}$$

conditional on that cloud. Equation (5) is not $\hat{\pi}_{\text{enkf},t}$. A comparison of $\pi_{\text{knn},t}$ with π_t mixes the two errors in Figure 1.

In the linear Gaussian comparison of Section 4 the first k particles, in the current order, are a reference set and are always updated with Algorithm 1. Every other particle takes the inverse-distance prediction of the reference particles' incremental log-likelihoods and does not run a forecast for that weight. When a rejuvenation is triggered, the move screens with the surrogate built from the reference cloud, and a proposal that passes the screen is still accepted or rejected with the ensemble Kalman likelihood. The weight and the screen are therefore not the same object inside that run. The weight of a non-reference particle targets the surrogate. The move, on the proposals that reach the second stage, still uses $\hat{\pi}_{\text{enkf}}$. We report the weight error directly, by recomputing (3) at a subset of non-reference particles, rather than by treating the whole cloud as a draw from one of the two posteriors.

3.5 A bound that does not close

A uniform bound on the log-likelihood difference would control the two posteriors built from two fixed log-likelihoods. The following calculation is conditional on that hypothesis. It is not a theorem about π_{knn}.

Lemma 2. *Let ℓ_1 and ℓ_2 be fixed log-likelihoods, and define $\pi_i(\theta) \propto \pi(\theta)\exp(\ell_i(\theta))$ for $i = 1, 2$, with both normalising constants finite and positive. If $|\ell_1(\theta) - \ell_2(\theta)| \le \delta$ at every θ in the support of the prior, then*

$$\|\pi_1 - \pi_2\|_{\text{TV}} \le \min\bigl(1,\ (e^{2\delta} - 1)/2\bigr). \tag{6}$$

Proof. Set $r(\theta) = \exp(\ell_1(\theta) - \ell_2(\theta))$ and $\mu = \mathrm{E}_{\pi_2}[r]$. The hypothesis gives $e^{-\delta} \le r(\theta) \le e^{\delta}$, and the same bounds pass to μ. The density ratio $\rho = \pi_1/\pi_2 = r/\mu$ therefore satisfies $e^{-2\delta} \le \rho \le e^{2\delta}$. The larger deviation from 1 is $e^{2\delta} - 1$, since $1 - e^{-2\delta} \le e^{2\delta} - 1$ for $\delta \ge 0$. Total variation is half the expectation of $|\rho - 1|$ under π_2, and it is at most 1, which is (6). □

The hypothesis was not established for $\ell_{\text{knn},t}$ and the ensemble Kalman log-likelihood. Three obstructions are enough to stop the argument.

Firstly, both log-likelihoods in Lemma 2 are fixed functions. The ensemble Kalman log-likelihood is a random function of the ensemble. The surrogate log-likelihood is a function of whichever cloud has been stored. A fresh ensemble Kalman evaluation, of the kind used in the diagnostic, is a new draw. The marginal $\hat{\pi}_{\text{enkf}}$ still contains that randomness through the extended target. Equation (5) conditions on the stored cloud. The two objects in the lemma are not these two targets.

Secondly, the diagnostic in Section 4 is a median absolute difference of one increment, on eight non-reference particles, averaged over replicates. It is not a uniform δ on the log-likelihood of the series, over the support of the prior. Substituting the diagnostic into (6) would not prove a total-variation bound.

Thirdly, a Lipschitz route does not supply the missing δ. If a single fixed log-likelihood were Lipschitz with constant L, the absolute error of the inverse-distance average would be at most

$kL/\sum_{i=1}^{k} d_{[i]}^{-1}$: each neighbour contributes an error of at most $Ld_{[i]}$, and the weight $d_{[i]}^{-1}$ cancels that factor. No finite Lipschitz constant for the ensemble Kalman log-likelihood is available on the support of the gamma priors used below, and that support is unbounded. The same comparison does not apply across two independent ensemble draws.

Chopin et al. (2012, Proposition 1) does not fill the gap, because the nearest-neighbour weight is not unbiased for the ensemble Kalman likelihood. Sherlock et al. (2016, Section 5.1) warn that targeting the surrogate invalidates draws from the expensive posterior, and they do not bound the difference. Katzfuss et al. (2018, Proposition 2) concerns the variance of a log ensemble Kalman likelihood under independent forecast draws. Variance of that kind points away from a small uniform δ: the log-likelihood moves when the ensemble is redrawn. The diagnostic in Section 4 remains the comparison we report. We do not state a theorem for the distance between π_{knn} and $\hat{\pi}_{\mathrm{enkf}}$.

4 Simulation studies

Two comparisons are reported. The first is linear and Gaussian. It checks Lemma 1, illustrates the Rao-Blackwellised weight when the observation variance inside the filter is wrong, and measures the nearest-neighbour weight of Section 3.4. The second is the Lotka–Volterra comparison of the plain filter, the Rao-Blackwellised filter and bootstrap SMC2. All algorithms were coded in R. The Lotka–Volterra runs used R 4.3.3 on an Intel Xeon processor.

The linear Gaussian runs use fewer parameter particles, a shorter series and a shorter gold-standard chain than Section 5.1 of Golightly et al. (2026). The Lotka–Volterra runs use fewer parameter particles, a fixed ensemble size and fewer replicates than Section 5.2 of that paper, and they do not use the particle Markov chain Monte Carlo reference used there. The numbers below are not a reproduction of those tables. Settings that the source paper leaves unspecified are stated here and were fixed before the runs.

4.1 Linear Gaussian checks

The latent process is the scalar Ornstein–Uhlenbeck diffusion of Section 5.1 of Golightly et al. (2026),

$$\mathrm{d}X_t = \theta_1(\theta_2 - X_t)\,\mathrm{d}t + \theta_3\,\mathrm{d}W_t, \tag{7}$$

with exact Gaussian transition. Over a step of length Δt the mean is $xe^{-\theta_1\Delta t}+\theta_2(1-e^{-\theta_1\Delta t})$ and the variance is $\theta_3^2(1-e^{-2\theta_1\Delta t})/(2\theta_1)$. We take $\Delta t = 1$, truth $\theta = (1,2,1)^\top$ and $x_0 = 10$. Sixteen observations are generated at times $1,\ldots,16$, with $H = 1$ and $R = 0.1$. There is no observation at time 0. The initial condition and R are treated as known. The prior is independent, $\theta_1 \sim$ Gamma$(2,2)$, $\theta_2 \sim$ Gamma$(5,3)$ and $\theta_3 \sim$ Gamma$(2,5)$, in the shape-rate parameterisation. Golightly et al. (2026) write Gamma(a,b) and do not name the parameterisation. We use the same reading throughout.

Each filter uses $M = 60$ parameter particles, a fixed ensemble of size $N = 12$ and $\gamma = 0.4$. Four replicate series are used. The reference on each series is a random-walk Metropolis–Hastings chain on $\log\theta$, targeting the exact Kalman likelihood times the prior, with independent proposal standard deviation 0.15 on each log-coordinate. The chain has 3000 iterations after a burn-in of 1000. The posterior mean of $\log\theta_i$ from the post-burn-in sample is the reference value. This chain is much shorter than the run of 10^6 iterations in Section 5.1 of Golightly et al. (2026), and its Monte Carlo error remains in every absolute error below.

The reported error is the mean, over the four series and over the three log-parameters, of the absolute difference between the terminal weighted posterior mean and the reference value. Transition draws count model integrations.

Identity. When the observation variance inside the filter equals the observation variance of the data, $R_g = R = 0.1$, the log ensemble Kalman increment and the log Rao-Blackwellised increment agree. The maximum absolute difference, over particles and times, is 0 on each of the four replicates. Lemma 1 is the corresponding statement. The correction does not change the weight, and the extra density evaluations have nothing to correct. The mean absolute error of this run against the short reference chain is 0.121, from 31248 transition draws. That figure is the error of the ensemble Kalman posterior at this particle count. It is not an error of the Rao-Blackwellised correction.

Wrong observation variance. The data and the reference chain still use $R = 0.1$. The filter uses $g = \mathcal{N}(y; x, 0.5)$, so f and g are both linear and Gaussian and they are not equal. The plain filter, which targets $\hat{\pi}_{\mathrm{enkf}}$ under this g, has mean absolute error 0.369 from 22776 transition draws. The Rao-Blackwellised filter, which uses the same g and then multiplies by f/g, has mean absolute error 0.169 from 36069 transition draws. The correction reduces the error in the posterior means and spends additional transition draws to do so. The ordering is an illustration on this series. It is not a general rule, and it is not the nonlinear observation of Section 4.2. Panel (b) of Figure 3 shows the two points.

Surrogate weight. The reference set has $k = 3$ particles, and the neighbour metric is Euclidean distance on $\log \theta$ after each coordinate has been scaled by the standard deviation of that coordinate in the stored cloud. A coordinate with standard deviation 0 is left unscaled. At the terminal time, (3) is recomputed for eight non-reference particles. The surrogate increment was to be called a poor approximation of the ensemble Kalman increment if the mean, over the four series, of the median absolute log difference exceeded 1. That threshold means a typical stored comparison differs by more than a factor of e. The mean of the median absolute log differences is 0.277. The diagnostic stays inside the threshold. It is not a uniform bound on the log-likelihood, and it is not substituted into (6).

Table 2 and panel (a) of Figure 3 compare the surrogate weight with two filters that target $\hat{\pi}_{\mathrm{enkf}}$. The plain filter uses a Metropolis–Hastings rejuvenation and the ensemble Kalman weight. Delayed acceptance uses the neighbour average as a screen only, with the same $k = 3$, and evaluates the ensemble Kalman likelihood at the second stage. The surrogate-weight run uses the construction in Section 3.4: non-reference weights are predictions, and the move still corrects the screen. The mean absolute errors are 0.172, 0.126 and 0.293, from 33855, 23970 and 5217 transition draws. The two schemes that target $\hat{\pi}_{\mathrm{enkf}}$ have the smaller errors. The surrogate weight uses fewer transition draws than either of them, and its error is larger. The drop from 33855 draws to 23970 is the screen, which leaves $\hat{\pi}_{\mathrm{enkf}}$ invariant. The further drop to 5217 is the replacement of most incremental weights by the surrogate, which is the second error.

Table 2: Ornstein–Uhlenbeck series. Mean absolute error of the terminal weighted posterior means of $\log \theta_i$, and mean number of transition draws, over four replicate series. The plain filter and delayed acceptance target $\hat{\pi}_{\mathrm{enkf}}$. The surrogate weight targets π_{knn} on the non-reference particles.

Scheme	Mean absolute error	Transition draws
Plain filter	0.172	33855
Delayed acceptance	0.126	23970
Surrogate weight	0.293	5217

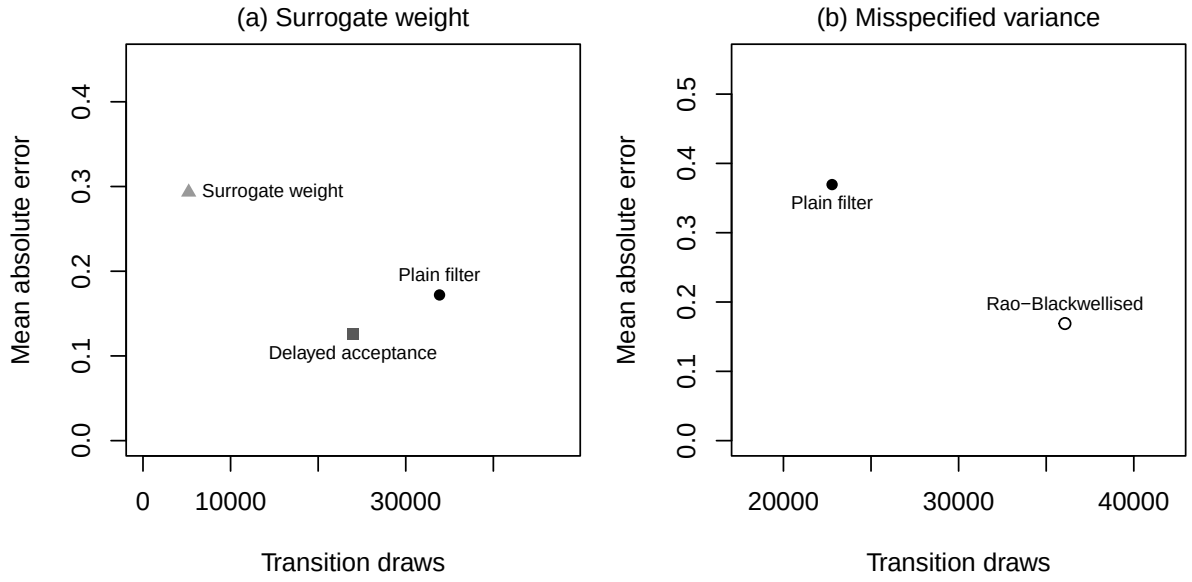


Figure 3: Ornstein–Uhlenbeck series. Mean absolute error against mean transition draws. (a) Three schemes on the correctly specified observation variance. The plain filter and delayed acceptance target $\hat{\pi}_{\text{enkf}}$. The surrogate weight does not. (b) Plain and Rao-Blackwellised filters when the observation variance inside the filter is 0.5 and the data use variance 0.1.

4.2 Lotka–Volterra targets

The latent process is the prey–predator diffusion (8) of Golightly et al. (2026, Section 5.2),

$$\mathrm{d}X_t = a(X_t, \theta)\,\mathrm{d}t + b(X_t, \theta)^{1/2}\,\mathrm{d}W_t, \tag{8}$$

with drift

$$a(x, \theta) = \big(\theta_1 x_1 - \theta_2 x_1 x_2,\ \theta_2 x_1 x_2 - \theta_3 x_2\big) \tag{9}$$

and diffusion matrix

$$b(x, \theta) = \begin{pmatrix} \theta_1 x_1 + \theta_2 x_1 x_2 & -\theta_2 x_1 x_2 \\ -\theta_2 x_1 x_2 & \theta_2 x_1 x_2 + \theta_3 x_2 \end{pmatrix}. \tag{10}$$

The observation model is Gaussian with mean and variance both equal to the prey component, and the predator component is latent. In the notation of (2), $H = (1, 0)$ and the variance is Hx_t rather than a constant matrix. Figure 2 is this update. The truth is $x_0 = (50, 50)^\top$ and $\theta = (0.5, 0.0025, 0.3)^\top$. The prior is independent, $\theta_1 \sim \text{Gamma}(2, 4)$, $\theta_2 \sim \text{Gamma}(20, 10^4)$ and $\theta_3 \sim \text{Gamma}(2, 4)$, again in the shape-rate parameterisation.

The synthetic path uses Euler–Maruyama with time step 10^{-3} on $[0, 40]$ and is thinned by 2000. Twenty prey observations are kept, at times $2, 4, \ldots, 40$. The initial condition is treated as known and is not an observation. Inside each filter the transition between observations is ten Euler steps of length 0.2. The diffusion matrix (10) is a covariance for positive states. After every Euler draw, and after every ensemble Kalman analysis update, a coordinate below 10^{-6} is replaced by 10^{-6}. Golightly et al. (2026) do not state a boundary rule.

Three filters are run on this one series, with four algorithm replicates. The plain nested filter uses $\tilde{R} = H\hat{\mu}_{t|t-1}$ and targets $\hat{\pi}_{\text{enkf}}$. The Rao-Blackwellised filter multiplies that variance by 2, evaluates (4), and targets $\tilde{\pi}_{\text{enkf}}$. Bootstrap SMC2 uses the Euler–Maruyama transition as its proposal and the observation density as its state weight. For that discretisation its parameter target is π_t. Each filter uses $M = 60$ and $\gamma = 0.4$. The ensemble size is fixed at $N = 12$ for the first two filters. SMC2 uses 24 state particles. There is no delayed acceptance and no growth of N. The rejuvenation proposal is one random walk, with $\zeta^2 = 2.56^2/3$. Section 5.2 of Golightly

et al. (2026) used $M = 1000$, an ensemble that starts at 20 and then grows, 50 replicates, delayed acceptance with three neighbours, and a particle Markov chain Monte Carlo reference of 10^6 iterations with 100 state particles. None of those settings is used here. SMC2 is the reference for π_t, and it is a noisy one at this state-particle count.

The reported summary is the terminal weighted mean of $\log \theta_j$. Table 3 gives the average of that summary over the four replicates, with the replicate standard deviation in parentheses. Figure 4 plots the difference of each nested filter from SMC2.

Table 3: Lotka–Volterra comparison. Mean, over four algorithm replicates, of the terminal weighted posterior mean of $\log \theta_j$, with the replicate standard deviation in parentheses. Draws are the mean number of Euler steps. Effective sample size is the mean terminal effective sample size.

Filter	$\log \theta_1$	$\log \theta_2$	$\log \theta_3$	ESS	Draws
Plain	-0.714 (0.122)	-6.035 (0.104)	-1.276 (0.101)	33.3	448500
Rao-Blackwellised	-0.852 (0.084)	-5.971 (0.159)	-1.196 (0.156)	42.6	476100
SMC2	-0.837 (0.064)	-5.977 (0.138)	-1.241 (0.161)	43.5	905700

The comparison was fixed before the runs. Let $m(a, j)$ be the average, over the four replicates, of the terminal weighted mean of $\log \theta_j$ under filter a, and let $s(a, j)$ be the sample standard deviation of those four means, with divisor $n-1$. The standard error of $m(\text{plain}, j) - m(\text{SMC2}, j)$ is

$$\sqrt{\big(s(\text{plain}, j)^2 + s(\text{SMC2}, j)^2\big)/4}. \tag{11}$$

Table 2 of Golightly et al. (2026) reports estimators of $\mathrm{E}(\log \theta_i)$. The prose of their Section 5.2 says that the plain filter shifts the posterior means of θ_2 and θ_3 downward. We apply the comparison to $\log \theta_2$ and $\log \theta_3$, which is the scale of that table. The downward shift is resolved on a coordinate if the plain-filter difference is negative and its absolute value exceeds (11). It is reported as appearing only if it is resolved on both coordinates, and as failing to appear only if the plain-filter mean is at least the SMC2 mean on both. Otherwise it is unresolved.

For $\log \theta_2$ the difference is -0.0586, with standard error 0.0865. For $\log \theta_3$ the difference is -0.0354, with standard error 0.0953. Both differences are negative, so the point estimates lie in the direction reported by Golightly et al. (2026), and neither absolute difference exceeds its standard error. The shift is unresolved. The bars in Figure 4 include zero on both coordinates. The Rao-Blackwellised mean is closer to the SMC2 mean on $\log \theta_2$, where the absolute gap is 0.0055 against 0.0586, and further from it on $\log \theta_3$, where the absolute gap is 0.0446 against 0.0354. A reduction on both coordinates does not hold.

For $\log \theta_1$, which was not part of the shift comparison, the plain-filter mean exceeds the SMC2 mean by 0.1227, with standard error 0.0688. The absolute difference exceeds the standard error, and the sign is positive. The Rao-Blackwellised gap on that coordinate is 0.0152. The mean number of proposals is 405 in each filter, and the mean numbers of acceptances are 79.25, 76.25 and 83.25. The largest fraction of Euler draws that met the floor is 0.008064. The draw counts in Table 3 do not include the observation-density evaluations inside (4), so they understate the cost of the Rao-Blackwellised weight relative to the plain filter.

5 Discussion

We have separated two errors that a nested ensemble Kalman filter can make, and we have kept the posteriors they induce distinct from the exact posterior and from each other. Golightly et al. (2026) already define the plain filter and the Rao-Blackwellised filter, compare them with SMC2 on a Lotka–Volterra diffusion, and use a nearest-neighbour average as a screen inside delayed acceptance. Their discussion suggests the further step of using that average as the weight. The

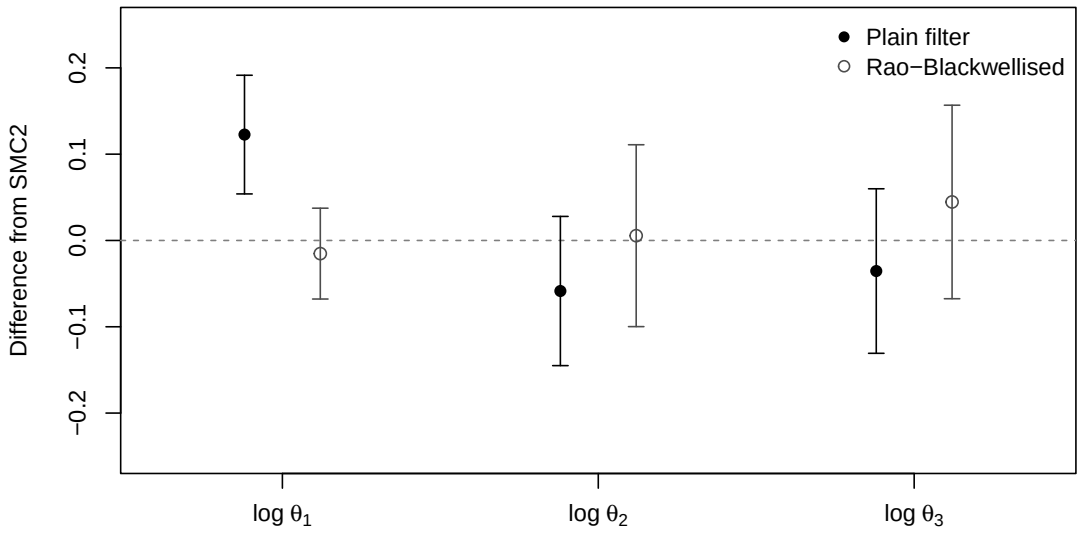


Figure 4: Lotka–Volterra series. Difference, from the SMC2 mean, of the average terminal weighted posterior mean of $\log \theta_j$. The bars are the standard errors (11), using the Rao-Blackwellised replicate standard deviation in place of the plain-filter standard deviation for the open circles. The dashed line is equality with SMC2. The shift comparison uses $\log \theta_2$ and $\log \theta_3$.

present paper makes that separation. A screen corrected by the ensemble Kalman likelihood remains a sampler for $\hat{\pi}_{\mathrm{enkf}}$. A weight that is the surrogate is a sampler for π_{knn}. Scoring the second against SMC2, or against a gold-standard run for π_t, would attribute the surrogate error to the ensemble Kalman approximation. The unbiased correction of a plug-in Gaussian density, as in Drovandi et al. (2022) and Temfack and Wyse (2026), removes a different error again and does not restore π_t.

Lemma 1 is the linear Gaussian case of the Rao-Blackwellised weight. On the Ornstein–Uhlenbeck series the two log increments agreed exactly when the filter used the observation variance of the data, so the correction is not worth its cost in that case. The same series with the filter variance set to 0.5, against data generated at variance 0.1, gives the complementary illustration: the Rao-Blackwellised mean absolute error was 0.169 and the plain-filter error was 0.369. The correction can move the posterior means when f and g differ, and the identity says that this movement is not available when they agree. Golightly et al. (2026) compare the two filters on observations that are not linear and Gaussian. The identity supplies the case in which that comparison would be empty, which their applications do not isolate.

The surrogate diagnostic stayed inside the threshold fixed in advance. The mean of the median absolute log differences was 0.277, against a threshold of 1. On the same series the run that replaces non-reference weights by the surrogate had mean absolute error 0.293, against 0.172 for the plain filter and 0.126 for delayed acceptance, and it used 5217 transition draws against 33855 and 23970. The accuracy-cost comparison in Figure 3 is the content of the second error at this particle count: fewer forecasts, and a larger error in the posterior means, because the target has changed. Lemma 2 states the total-variation bound that would hold for two fixed log-likelihoods differing by at most δ. The ensemble Kalman log-likelihood is not a fixed function, the diagnostic is not a uniform δ, and no Lipschitz constant on the gamma prior was available. We therefore do not claim a distance between π_{knn} and $\hat{\pi}_{\mathrm{enkf}}$.

On the Lotka–Volterra series the plain filter's means of $\log \theta_2$ and $\log \theta_3$ lie below the SMC2 means, by 0.0586 and 0.0354. The standard errors are 0.0865 and 0.0953. Under the comparison

fixed before the runs, the shift is unresolved. Figure 4 shows the same fact: the standard-error bars on those two coordinates include zero. The Rao-Blackwellised filter is closer to SMC2 on $\log \theta_2$ and further from it on $\log \theta_3$. Golightly et al. (2026) report a downward shift for the plain filter, reduced by the Rao-Blackwellised weight, at $M = 1000$ and against a long particle Markov chain Monte Carlo run. Our comparison uses $M = 60$, four replicates and SMC2 itself as the reference. It can show a shift of the size in their Table 2, and it can miss it. Here it does not resolve it. The result does not confirm that ordering, and it does not contradict it. What it adds is a statement about the scale of the discrepancy relative to Monte Carlo error when the ensemble is small and the reference is SMC2, which their table does not report.

The limits of the Lotka–Volterra comparison are those of the study itself. There is one series. The state-particle count in SMC2 is fixed at 24, and a particle Markov chain Monte Carlo reference was not run. The standard errors are estimated from four replicates. A small fraction of Euler draws met the positive floor. The Euler counts omit the cost of the density ratio in the Rao-Blackwellised weight. The gamma priors use a shape-rate reading that Golightly et al. (2026) do not state. These are reasons to treat the unresolved verdict as a statement about this comparison. The linear Gaussian gold-standard chains are likewise short, so the mean absolute errors in Section 4.1 are relative to those chains.

We anticipate that the separation is of most use where a practitioner is choosing among the plain filter, the Rao-Blackwellised filter and a surrogate weight, and would otherwise read every discrepancy from π_t as a defect of the ensemble Kalman likelihood. The correction is the wrong expense when the observation model is already linear and Gaussian. It is a relevant expense when a state-dependent variance is replaced by a function of the forecast mean, which is the Lotka–Volterra observation, whether or not a particular finite-M comparison resolves the shift. The surrogate weight is a different estimator, of a different posterior, and the screen is not a substitute for it.

A bound on π_{knn} that can be checked, for a likelihood that is redrawn with the ensemble, remains open. So does a repeat of the Lotka–Volterra comparison at the particle counts of Section 5.2 of Golightly et al. (2026). Gradient proposals inside the rejuvenation step, and a neighbour search whose cost scales more gently with the number of stored particles, are among the modifications suggested by Golightly et al. (2026). They change the move under a target that has already been chosen. They do not decide which of $\hat{\pi}_{\mathrm{enkf}}$, $\tilde{\pi}_{\mathrm{enkf}}$ and π_{knn} that target should be. We do not add an epidemic series. Temfack and Wyse (2026) already study a nested ensemble scheme for incidence.

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Data availability

All series used in this paper were simulated from the models in Section 4, with the priors, time grids and algorithm settings stated there.

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